1. Start Here: Ratios, Cents, and Why Strings Matter

The two ideas you need for everything else: pitch ratios and the cent

Pitch is a ratio, not a number

Two notes sound 'the same distance apart' whenever the ratio of their frequencies is the same. Going from 200 Hz to 300 Hz feels like going from 400 Hz to 600 Hz: both are a ratio of 3:2. Pitch perception is logarithmic, just like decibels.

The cent: a logarithmic ruler

cents = 1200 × log2(f2 / f1). An octave (2:1) is exactly 1200 cents, so one equal-tempered piano semitone is 100 cents. Trained ears hear about 5 cents; our measurements across 13 masters agree to a few cents, so the cent is the right unit for comparing theory with performance.

Why do simple ratios sound good?

If two tones have a ratio like 3:2, their harmonics line up (every 3rd harmonic of the lower note hits every 2nd of the upper one), so there is no 'beating'. The simpler the ratio, the smoother the sound. Ancient theorists discovered this with strings, long before anyone could measure frequency.

Strings turn ratios into lengths (this is the key for a setar player)

On a string at fixed tension, frequency is inversely proportional to the vibrating length. Press a fret at 3/4 of the string and the pitch rises by 4/3; at 2/3 it rises by 3/2; at 1/2 you get the octave. A fretted instrument is therefore a physical ratio calculator: every fret position is a theory statement.

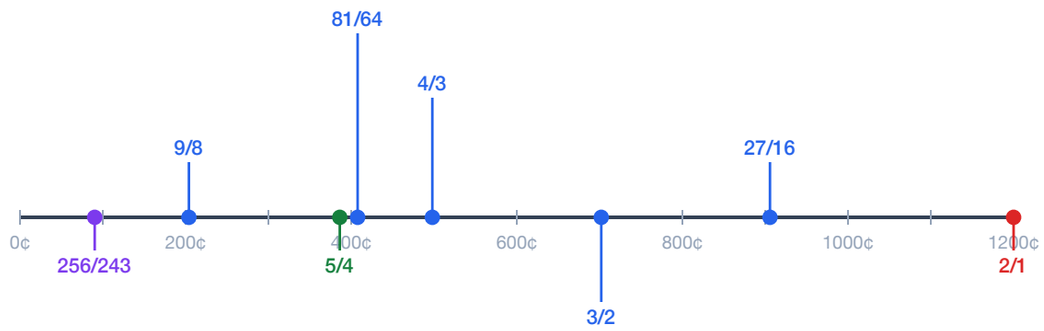
> Rule of thumb: ratio r above the open string → fret at 1/r of the length from the bridge, i.e. a distance (1 – 1/r) of the string from the nut. And stacking intervals means multiplying ratios (adding cents).

How this document is organised

- Page 2 Pythagoras: the 3-limit system that everything else builds on or reacts to.
- Page 3 The setar as a multiplication machine (frets × strings).
- Pages 4–7 Al-Farabi, Ibn Sina, Al-Urmawi, Al-Maraghi: one page each.
- Page 8 Modern 20th-century theories (Vaziri, Barkechli, Farhat).
- Page 9 Side-by-side comparison with our Mahur measurements. Page 10 timeline & glossary.

A cents ruler for the common intervals

0–1200¢; dots = pure-ratio intervals, ticks = 12-TET semitones every 100¢



Blue = Pythagorean (3-limit) · Green = 5/4 just third · Purple = limma

Note how 81/64 (408¢) and 5/4 (386¢) differ by 21.5¢ — a distinction that will matter later.

Ratio → cents → what 12-TET gives

The gaps between theory and the piano are the whole story

Interval	Ratio	Just cents	12-TET	Diff
Octave	2/1	1200.0	1200	+0.0
Fifth	3/2	702.0	700	+2.0
Fourth	4/3	498.0	500	−2.0
Major 3rd (Pyth.)	81/64	407.8	400	+7.8
Major 3rd (just)	5/4	386.3	400	−13.7
Minor 3rd (Pyth.)	32/27	294.1	300	−5.9
Whole tone	9/8	203.9	200	+3.9
Limma	256/243	90.2	100	−9.8

2. Pythagoras (c. 570–495 BCE) and the 3-Limit System

Everything from pure fifths: octave, fifth, fourth, whole tone, limma, comma

The idea

Tradition credits Pythagoras (Samos/Croton) with discovering that the consonances are small whole-number ratios on a monochord (a single stretched string with a movable bridge): octave 2:1, fifth 3:2, fourth 4:3. These use only the numbers 1–4 (the 'tetractys', 1+2+3+4 = 10), which the Pythagorean school considered sacred. The famous blacksmith-hammers story is a legend (it does not work physically), but the monochord facts are real. Written mathematics comes later (Philolaus, Plato's Timaeus, Euclid's Sectio Canonis).

Building a scale from just 2 and 3

- Whole tone = fifth − fourth = (3/2)/(4/3) = 9/8 = 203.9¢.
- Two stacked tones: 9/8 × 9/8 = 81/64 = 407.8¢ (the 'ditone', a bright major third).
- What is left in the fourth: (4/3)/(81/64) = 256/243 = 90.2¢, the limma ('leftover').
- So a Pythagorean diatonic tetrachord is tone–tone–limma: 204 + 204 + 90 = 498¢ (a pure fourth).

The catch: the Pythagorean comma

Stack twelve pure fifths and you land 23.46¢ sharp of seven octaves: 3^12 / 2^19 = 531441/524288 ≈ 1.0136. Pure fifths never close into a circle. That tiny leftover (the comma) is why every tuning system must make a compromise, and why Urmawi's 17-note idea (page 6) is interesting: he lets the chain run on instead of closing it.

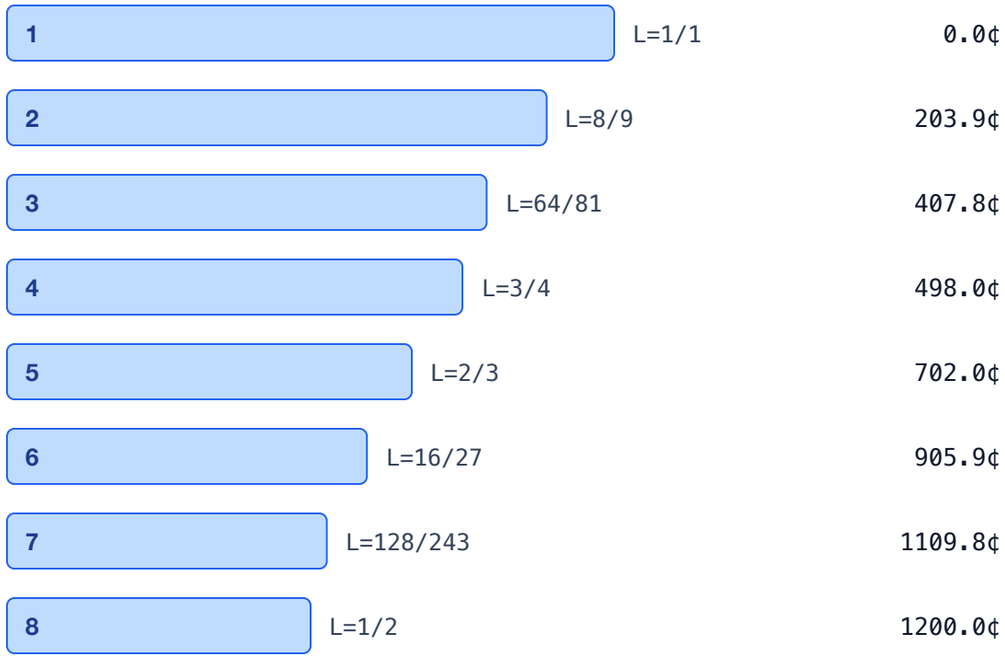
Why an engineer should care

Pythagorean tuning is the unique 'closed under multiplication by 3/2 and 2' set: every pitch is 2^a × 3^b. That makes it easy to compute, easy to tune by ear (just a chain of pure fifths), and perfect for fretted lutes where one fret serves several strings. Its weakness: thirds (81/64) are 21.5¢ wider than the 'sweet' 5/4 of just intonation.

> Everything later in this document is either an extension of this 3-limit system (Urmawi, Maraghi) or a reaction to it (Zalzal's neutral third, Vaziri's equal quarter-tones).

The monochord: one string, seven notes

Bar width = vibrating length (1 = full string). Shorter string → higher pitch.



Steps between bars: 9/8 · 9/8 · 256/243 · 9/8 · 9/8 · 9/8 · 256/243
(the Western major scale in Pythagorean tuning)

Chain of fifths: how the comma appears

Each step ×3/2, folded into one octave (cents above the start)



Fifth #12 (red) lands 23.5¢ above the starting note: the Pythagorean comma.
Numbers = how many fifths were stacked.

3. The Setar as a Multiplication Machine

How one fret serves two strings, and why this matters for the whole analysis

Pitch = open string × fret ratio

On a setar the open strings are tuned a fourth/fifth apart (we call the lower-register reference 'Do' and the string a fifth below it 'Sol↓'). Press a fret that raises the pitch by ratio r and the note is simply (open-string pitch) × r. In cents, you just add: cents(open) + cents(r).

The shared-fret trick

A fret is a single physical object tied across the whole neck. The same fret pressed on the Sol↓ string (pitch 3/4 of Do) gives a different note than on the Do string. So with a handful of frets a player gets two layers of pitches, and the second layer is not free: it is locked to the first by multiplication.

Example from our Mahur study. The 'Re' fret (9/8) pressed on the Sol↓ string (3/4) gives 9/8 × 3/4 = 27/32 = –294¢ (the note we label La↓). A 'free' just-intonation sixth would be 5/3 → –316¢; the instrument forces –294¢, and the masters' measured La↓ ≈ –300¢ confirms the frets, not an ideal ratio, decide the pitch.

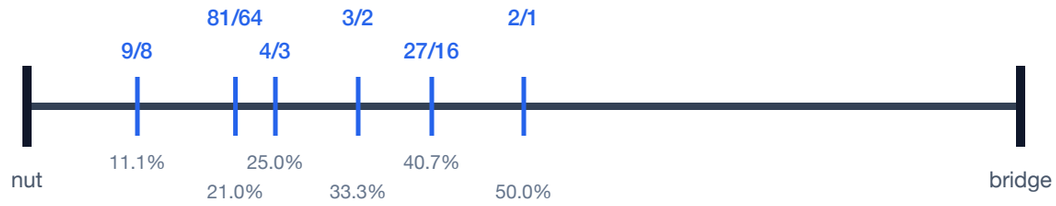
This is why measuring one fret on two strings is such a powerful test: the Pythagorean hybrid model predicts both numbers from one fret, with zero extra parameters.

Practical tuning facts (engineer cheat sheet)

- Fret distance from nut = L × (1 – 1/r). For r = 9/8 that is 11.1% of the string; for 4/3, 25%; for 3/2, 33%.
- A 1¢ error = 0.058% of frequency; on a ~600 mm string, 1¢ at the 4/3 fret is about 0.1 mm.
- Moving the 'Re' fret by 10¢ moves every note derived from it on the other string by the same 10¢.

Fret positions as fractions of string length

Distance from the nut (left) to the bridge (right); position = 1 – 1/r



One fret, two strings (cents relative to Do)

Do string = open pitch 0¢. Sol↓ string = open pitch –498¢ (ratio 3/4).

Fret ratio	On Do string	On Sol↓ string	Sol↓ ratio
1/1 (open)	+0.0¢	–498.0¢	3/4
256/243	+90.2¢	–407.8¢	64/81
9/8	+203.9¢	–294.1¢	27/32
32/27	+294.1¢	–203.9¢	8/9
81/64	+407.8¢	–90.2¢	243/256
4/3	+498.0¢	+0.0¢	1/1
3/2	+702.0¢	+203.9¢	9/8
27/16	+905.9¢	+407.8¢	81/64

Green numbers are the second layer produced for free by the same frets.

E.g. fret 9/8 on Sol↓ = 27/32 = –294¢ (our La↓); fret 32/27 on Sol↓ = 8/9 = –204¢ (Si↓).

4. Abu Nasr al-Farabi (c. 872–950): Frets, Zalzal, and the Neutral Third

Baghdad / Damascus · Kitab al-Musiqa al-Kabir ('The Great Book of Music')

Who and why he matters

Al-Farabi was a polymath (philosopher, logician, musician) born in Transoxiana and active in Baghdad, Aleppo and Damascus. His Kitab al-Musiqa al-Kabir is the first systematic study of the 'ud and tunbur (long-necked lute, an ancestor of the setar) fretboard. He did what an engineer would do: catalogue every fret position used in practice and calculate their ratios.

Key contributions

- Fret catalogue of the 'ud and the Baghdad and Khorasan tunburs, comparing the Pythagorean frets with those used by players.
- Documented the 'Zalzal' middle finger fret named after the 8th-century lutenist Mansur Zalzal: a neutral third near 27/22 ≈ 355¢, roughly halfway between minor (294¢) and major (408¢).
- Alongside it he lists the 'wusta of the Persians' (32/27 ≈ 294¢) and the Pythagorean 'binsir' (81/64 ≈ 408¢), so one hand position has three or four candidate middle-finger notes.
- Classified consonances (octave, fifth, fourth and their combinations) and the genera of the tetrachord, bringing Greek theory (Aristoxenus, Ptolemy, Euclid) into Arabic/Persian practice.
- Used 'super-particular' (epimoric) ratios (n+1)/n: whole tone 9/8 (204¢), minor tone 10/9 (182¢), semitone 16/15 (112¢), and microtonal neutral steps like 12/11 (151¢) and 13/12 (138.6¢) to express fret divisions.

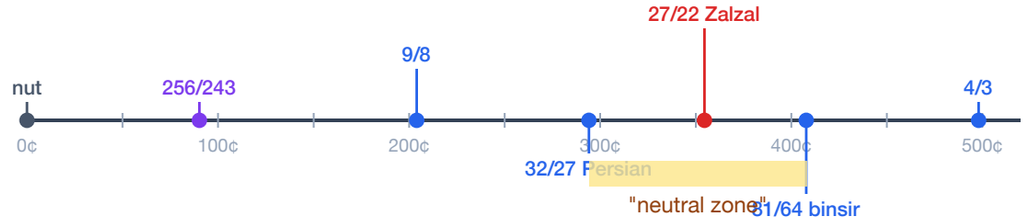
Why this matters for our data: where does 13/12 come from?

To a modern ear trained on 5-limit ratios (2, 3, 5), 13/12 looks bizarre. But on an instrument string, dividing the distance between fret frets into equal geometric parts produces consecutive superparticular steps: 11/10 (165¢) → 12/11 (151¢) → 13/12 (138.6¢) → 14/13 (128¢). Medieval theorists (Fārābī and later Ibn Sīnā) analyzed these as the natural mathematical home for neutral steps. When we measure our master recordings, Re^{k} sits right at +137.5¢ in Māhur and +138.7¢ in Chahārgāh — within 1¢ of 13/12, and 12¢ flat of Vaziri's artificial 24-TET 150¢!

Key takeaway: 13/12 is not an arbitrary modern curve-fit; it is the classical epimoric definition of a narrow neutral second (2/3-tone) documented in medieval lute fret analyses.

Farabi's middle-finger candidates on the 'ud

Cents above the open string; the neutral zone sits between the two Pythagorean thirds



Red = Zalzal neutral third (≈355¢) · Blue = 3-limit frets · Purple = limma
The Zalzal note is 60¢ above the Persian wusta and 53¢ below the binsir.

Timeline: who played what in the 'middle finger' slot

A thousand years of one finger

- **8th c.** Zalzal introduces the neutral middle fret on the 'ud
- **10th c.** Farabi catalogues Zalzal, Persian and Pythagorean frets
- **13th c.** Urmawi drops the neutral and closes the system on 3-limit (17 notes)
- **15th c.** Maraghi keeps Pythagorean frame; still describes older neutral usage
- **20th c.** Vaziri: equal quarter-tones; Barkechli/Farhat: measured neutral intervals

5. Ibn Sina / Avicenna (980–1037): Synthesis and the Persian-Language Treatise

Bukhara · Hamadan · Isfahan — music chapters of al-Shifa, al-Najat and Danishnama-yi 'Ala'i

Who and why he matters

Ibn Sina is best known as a physician and philosopher, but he wrote three major music chapters: in al-Shifa ('The Healing'), in al-Najat ('The Deliverance') and, in Persian, in Danishnama-yi 'Ala'i ('Book of Knowledge for 'Ala' al-Dawla'). The Persian version is notable: it moves music theory from Arabic into the language of the Persian court.

Key contributions

- Synthesis: reconciles the Greek mathematical tradition (Euclid, Aristoxenus, Ptolemy) with the practice-based Arab lute tradition of Farabi.
- Consonance theory: explains why intervals are consonant by ratio, ranks them and describes how larger ones are built from smaller ones.
- Genera of the tetrachord (jins): the diatonic, chromatic and enharmonic divisions of a fourth, including the soft 'equable' divisions that give neutral steps.
- Rhythm (iqa'): a formal treatment of rhythmic cycles that later became central in Urmawi's Adwar.
- Pitch vs. instrument: separates the abstract interval system from any one instrument's fretting.

A tetrachord is a 4:3 box split into three steps

The tetrachord (jins) is the building block: a perfect fourth (498¢) divided into three steps. Where you place the two inner notes defines the 'genus' (flavour). The diagram on the right compares four divisions: the Pythagorean diatonic, Ptolemy's intense diatonic, Ptolemy's equable diatonic (neutral steps) and a 24-TET division.

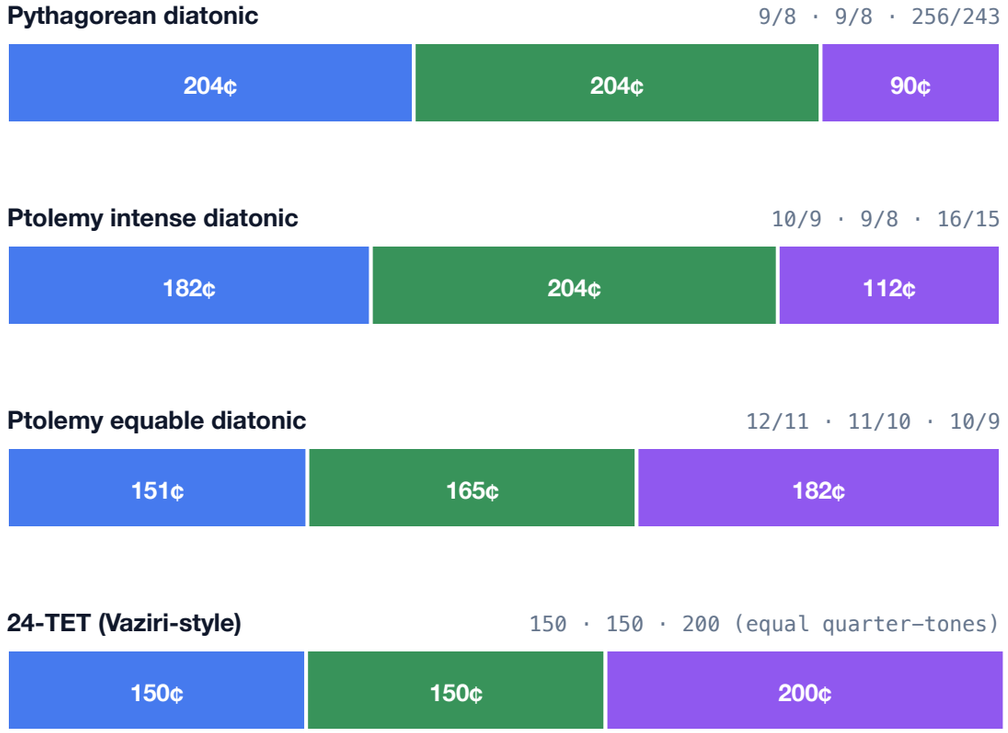
Why this matters for us

In every dastgah each jins is a miniature 4:3 box. Our measurements show the outer boundary of each box (Do–Fa, Do–Sol) is very precisely 4:3 / 3:2, but the inner notes are where the schools differ.

Ibn Sina's importance for this project is conceptual: he fixes 'tetrachord = fixed outer fourth + flexible inner notes', which is exactly the structure we see empirically.

Four ways to split a perfect fourth (498¢)

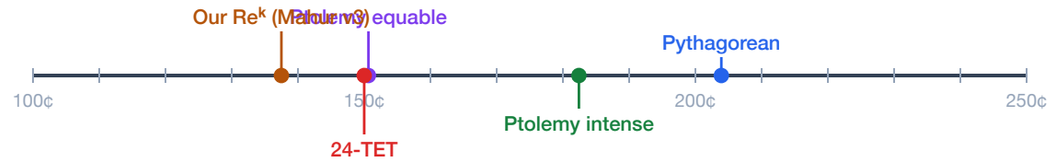
Bar segments = the three steps of the tetrachord, in cents



All four add up to 498¢; only the two inner notes move.

What a neutral step looks like

Second note of each tetrachord, cents above the first



First step of the tetrachord: our measured Re^k (≈137.5¢) sits with the small neutral steps, not with the 204¢ Pythagorean tone and not exactly on the 24-TET 150¢.

6. Safi al-Din al-Urmawi (c. 1216–1294): The 17-Note Pythagorean Octave

Urmia → Baghdad · Kitab al-Adwar (c. 1250s) and al-Risala al-Sharafiyya (c. 1267) — the 'Systematist school'

Who and why he matters

Urmawi was a scribe, singer, lutenist and theorist at the court of the last Abbasid caliph in Baghdad; he survived the Mongol conquest (1258) and served the Ilkhanid vizier Juvayni, to whom the Sharafiyya is dedicated. His books are the foundation of what historians call the 'Systematist school' of Persian/Arabic music theory.

Key contributions

- 17 notes per octave from a chain of pure fifths. Instead of closing the circle at 12, he lets the chain run to 17, giving two kinds of small step: the limma (90.2¢) and the Pythagorean comma (23.5¢).
- Fretting the 'ud so that all 17 notes of the octave can be found, replacing the ambiguous neutral frets of the earlier tradition.
- The mode system: 12 maqamat plus 6 avazat — the ancestor of our dastgah/avaz organisation.
- A rhythmic theory of cycles (adwar) and the first precise written notation using letters for pitch and rhythm.
- Your uploaded diagram: the double-octave drawn as 4:3 × 4:3 × 9:8 arches (2:1 = 4:3 × 4:3 × 9:8) from his manuscript tradition. Pure 4:3 fourths and a 9/8 tone are the whole skeleton.

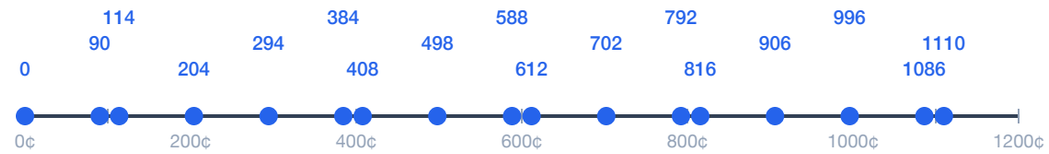
What this means in our data

The outer skeleton we measure in Mahur (Sol↑ +702.1¢, Fa↑ +495.9¢, Re +201.5¢) matches the 3-limit frame within 0.1–2.4¢. The only deviation from Urmawi's frame is the major third: Nariman and Saba play 5/4 (+386¢), the 'just' third that appears as an alternative table in the Sharafiyya manuscript.

> Urmawi's lasting trick: treat the 17 notes as a menu. A given mode only uses 7–8 of them, and different modes pick different subsets of the same 17 positions.

The 17 notes of the octave (folded chain of fifths, k = –8…+8)

Cents above the tonic; tick spacing shows limma vs comma steps



Step sizes between neighbours: 23.5¢, 90.2¢

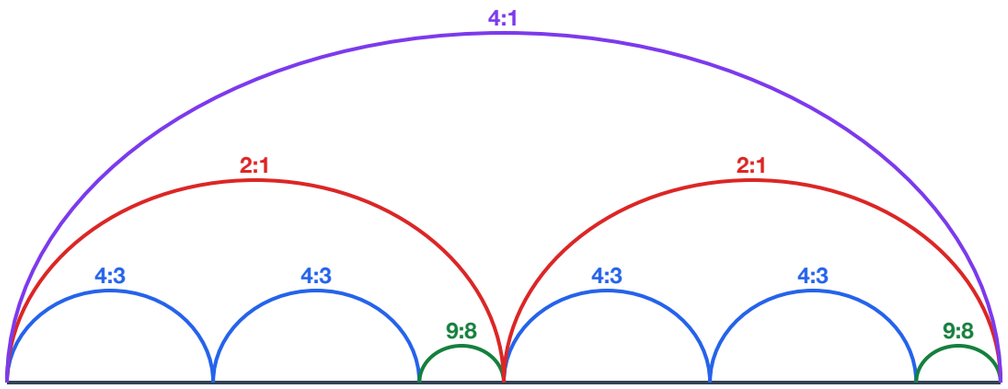
Limma = 90.2¢ (256/243) · Comma = 23.5¢ · Apotome = 113.7¢ (limma + comma)

Whole tone 203.9¢ = limma + limma + comma (90.2 + 90.2 + 23.5)

Chain extended to 17 fifths yields only two distinct step sizes: limma and comma.

The double-octave skeleton from the manuscript

2:1 = 4:3 × 4:3 × 9:8, drawn as arches



0¢ … 498¢ … 996¢ … 1200¢ … 1698¢ … 2196¢ … 2400¢ (two octaves)

7. Abd al-Qadir al-Maraghi (c. 1353–1435): Persian-Language Systematist

Maragha → Tabriz → Baghdad → Samarkand → Herat · Jami' al-Alhan (1405), Maqasid al-Alhan (c. 1418)

Who and why he matters

Maraghi was court musician under the Jalayirids and then the Timurids (Shahrukh in Herat). He is the great heir of Urmawi, but wrote in Persian rather than Arabic, which made his work the standard for later Persian musicians.

Key contributions

- Kept and refined Urmawi's 3-limit frame: pure fifths, fourths and 9/8 tones, building every step from them.
- His preferred minor third is the Pythagorean 32/27 = 294.1¢ and major third 81/64 = 407.8¢ — both are 'frets in the 3-limit chain', not neutral pitches.
- Practical, performance-oriented theory: modes (maqam), shu'ab (offshoots) and complete compositions (naqsh, qawl, tarana, ghazal), so theory is tied to a living repertoire.
- Described the lute (ud) and its 'Persian' tunings so that fretted instruments could realise the whole system.
- Bridge to the Safavid period: his Persian terminology survives in modern dastgah vocabulary.

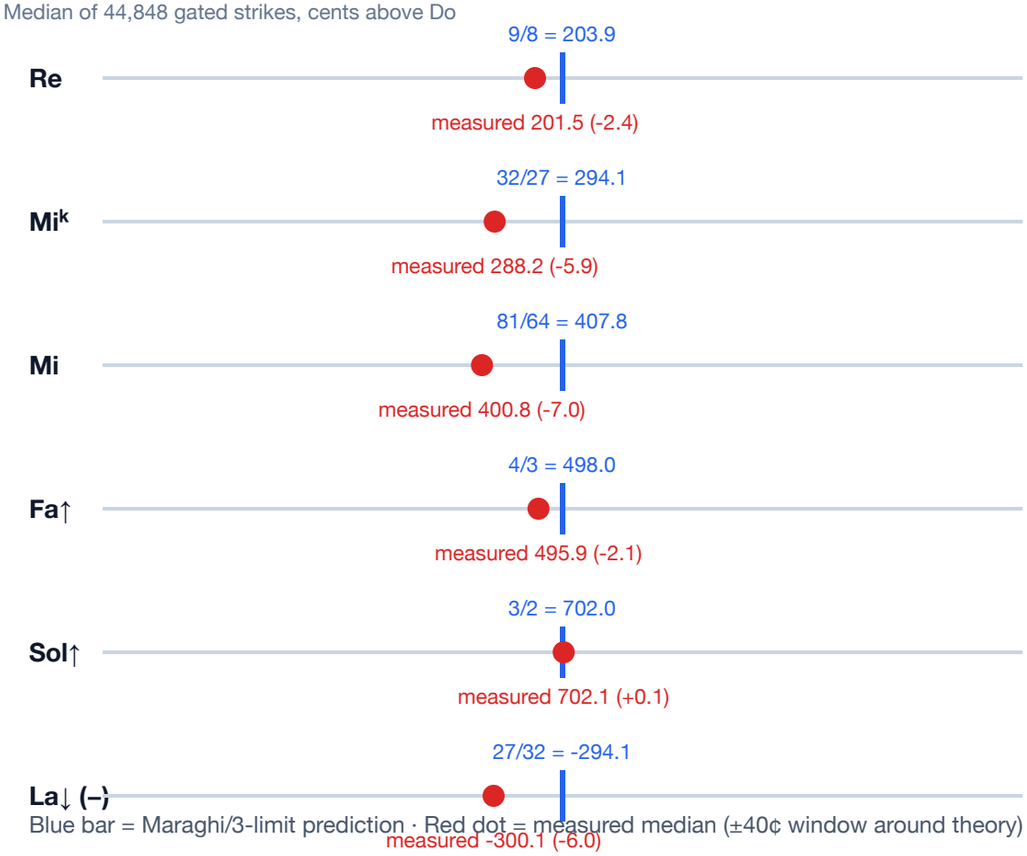
Why this matters for our data

Our modern masters' Mi↓ (Mi^{k} in Shekaste) ≈ +288.2¢ is just 5.9¢ below Maraghi's 32/27 (294.1¢) and nowhere near the old Zalzalian ≈340¢; the major third Mi ≈ +400.8¢ sits between his 81/64 (407.8¢) and Urmawi's 5/4 (386.3¢). The Qajar-trained masters (Borumand, Forutan, Neydavud) are the exception: their Mi^{k} stays near 332–341¢.

Net result: modern Persian performance is closer to Maraghi's 3-limit frame than to Vaziri's equal-tempered quarter-tones, with two clear departures: the major third (scatter) and the old neutral third (survival).

Maraghi's frame vs. what the masters actually play (Mahur v3)

Median of 44,848 gated strikes, cents above Do



Blue bar = Maraghi/3-limit prediction · Red dot = measured median (±40¢ window around theory)

Same-fret check: Mi↓ on Do and Si↓ on Sol↓

One fret ⇒ two predictions

Fret 32/27 on Do string → Mi↓ (Mi^{k})

theory +294.1¢ measured +288.2¢ Δ −5.9¢

Same fret on Sol↓ string → Si↓

theory −203.9¢ measured −211.0¢ Δ −7.1¢

Both come within ~6¢ of the Pythagorean prediction — the modern single fret is shared.

Ali-Naqi Vaziri (1887–1981): 24 equal quarter-tones

Vaziri, who studied in Berlin, published Dastur-e Tar (1913) and Musiqi-ye Nazari (1934). He proposed dividing the octave into 24 equal quarter-tones (24-TET) and introduced the signs koron (a quarter-tone flat) and sori (a quarter-tone sharp). Pro: simple, notatable, transposable, great for pianos and notation software. Con: it forces every neutral interval to exactly 150¢ / 350¢ and ignores the pure fifths and fourths real performers play.

Mehdi Barkechli (1908–1990): Pythagorean, with limma and comma

In Les systèmes de la musique traditionnelle de l'Iran (1963), written with Nur-Ali Borumand's radif as source, Barkechli argued that the intermediate pitches come from the Pythagorean system itself: limma (≈90¢) and comma (≈24¢) steps. It brings Persian theory back to the medieval 3-limit line, but it makes koron a small, fixed step.

Hormoz Farhat (1929–2021): measure, don't assume

In The Dastgah Concept in Persian Music (1990) Farhat recorded performers on the tar and analysed the scale degrees. His central finding: the tuning is not equal-tempered, the fourth and fifth are close to pure, and the neutral steps are flexible, with no single quarter-tone shared by every player. He is the closest precedent for what we are doing.

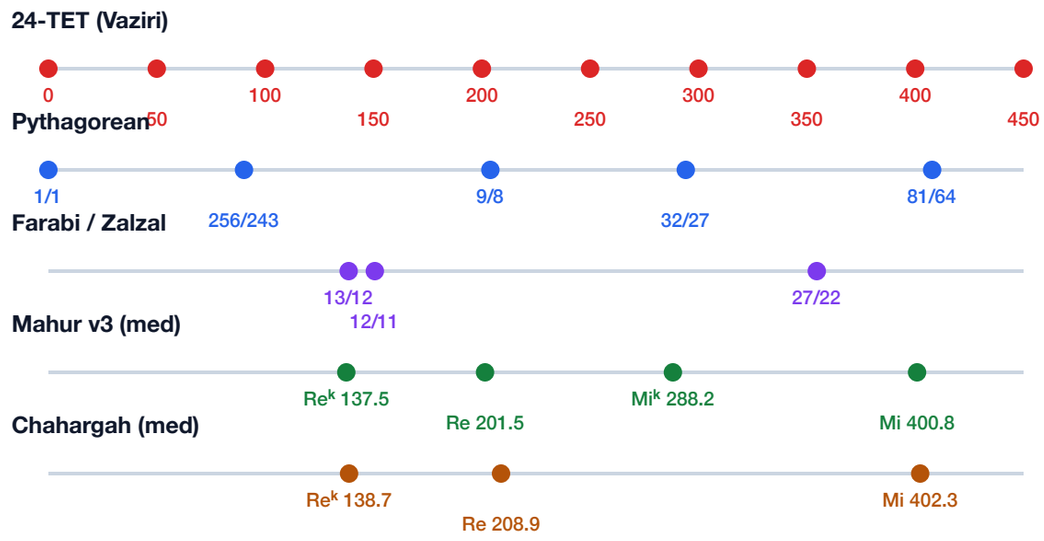
Where does our 13-master study sit? (Māhur & Chahārgāh)

- Fifths/fourths: pure within 0.1–2.4¢ — against Vaziri (his fifth is 700¢, 2¢ flat), with Barkechli, Urmawi and Maraghi.
- Koron Re^k is invariant across Dastgāhs: +137.5¢ in Māhur, +138.7¢ in Chahārgāh! Exactly matching 13/12 (138.6¢), and rejecting Vaziri (150¢) and Barkechli (90¢).
- Mi^k: two populations (≈288¢ modern, ≈335¢ Qajar-trained): a clear historical tuning shift that no single static theory predicts.

Cross-Dastgāh invariance: whether Re^k is an incidental passing note (Māhur Delkash) or the primary structural pillar (Chahārgāh Darāmad/Zābol), the masters tune it to the identical ~138¢ fret position with sub-2¢ precision!

Zoomed ruler 0–450¢: four models vs measured notes

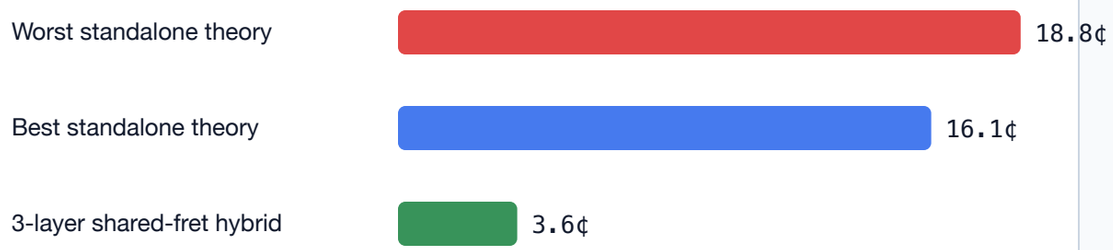
Rows: 24-TET · Pythagorean · Zalzal/Farabi · Mahur v3 · Chahargah



Green dots sit on the blue/purple rows, not on the red 24-TET grid.

Scorecard: how well each model fits (Mahur v3)

RMS error across the 17 measured slots



Figures from page 5 of mahur_pitch_analysis.pdf: standalone theories scored 16.1–18.8¢ RMS. Notice how Re^k in Chahārgāh (+138.7¢) replicates Māhur (+137.5¢) within 1.2¢!

9. Side-by-Side: Who Said What About Each Note

Cents above Do / Tonic. Comparing historical theories against measured Māhur (44.8k), Chahārgāh (32.9k), & Dashti (10.3k strikes)

Note / Degree	Pythagoras / Urmawi	Farabi (Zalzal)	Vaziri 24-TET	Māhur v3 Measured	Chahārgāh Measured	Dashti Measured	Best Match & Analysis
Zir-e Ist / Subtonic	−203.9 (9/8)	—	−200	−204.2	**−206.1**	**−209.8**	**Pythagorean 9/8 subtonic**
Tonic / Ist (0¢)	0.0 (1/1)	0.0	0	0.0	**0.0**	**0.0**	**Acoustic center of gravity**
Koron 2nd (Re^{k}/Mi^{k})	—	138.6 (13/12)*	150	137.5	**138.7**	**135.8 / 137.4**	**Farabi 13/12 invariant across all 3!**
Whole Tone / 2nd	203.9 (9/8)	203.9	200	201.5	**208.9**	**—**	**Pythagorean 9/8 tone**
Minor 3rd (Fa / Mi^{k})	294.1 (32/27)	354.5 (27/22)	300	288.2 mod ~335 Qajar	**—**	**288.5**	**Maraghi 32/27 (mod) / Zalzal (Qajar)**
Major 3rd (Mi)	386.3 / 407.8	407.8 (binsir)	400	400.8	**402.3**	**—**	**Bridges Urmawi 5/4 & Maraghi 81/64**
Fourth (Fa / Sol)	498.0 (4/3)	498.0	500	495.9	**494.5**	**495.7**	**Pythagorean pure fourth (4/3)**
Motaghayyer (koron)	—	636.6 (Farabi)	650	—	**—**	**658.9**	**Neutral 5th in Dashti Dāng**
Fifth (Sol / 3/2)	702.0 (3/2)	702.0	700	702.1	**703.3**	**704.5**	**Pythagorean pure fifth (3/2)**

* Ratio attribution to Farabi is indicative; to be verified against the primary text (see page 4). "—" = not present or not structural.

Notice the universal invariance of the Koron 2nd: 137.5¢ in Māhur, 138.7¢ in Chahārgāh, 137.4¢ in fretless Dashti — all locking into Farabi 13/12 (138.6¢)!

10. Dariush Tala'i's Theory of Dāngs & Setār Neck Architecture

Visualizing the 4 structural tetrachords with pure mathematical ratios, historical tuning shifts, and the 5th superimposed string ruler

Tetrachords as Generative Grammar

Dariush Tala'i (1993, 2000) demonstrated that the Persian modal system is generated not from octaves, but from 4-note building blocks called dāngs (tetrachords spanning a pure fourth, 4/3 = 498.0¢).

Every dastgāh is built from four distinct dāng archetypes with precise mathematical ratios:

- 1. Dāng-e Shūr (N-N-T): 1/1 (0¢) → 13/12 (138.6¢) → 32/27 (294.1¢) → 4/3 (498.0¢). Core of Shūr, Dashti, and Navā.
- 2. Dāng-e Māhūr (T-T-S): 1/1 (0¢) → 9/8 (203.9¢) → 81/64 (407.8¢) / 5/4 (386.3¢) → 4/3 (498.0¢). Core of Māhūr.
- 3. Dāng-e Chahārgāh (N-A-S): 1/1 (0¢) → 13/12 (138.6¢) → 81/64 (407.8¢) → 4/3 (498.0¢). Symmetrical augmented-second dāng.
- 4. Dāng-e Esfahān / Minor (T-S-T): 1/1 (0¢) → 9/8 (203.9¢) → 32/27 (294.1¢) → 4/3 (498.0¢). Core of Bayāt-e Esfahān.

Historical Tuning Evolution on the Frets

- Neutral Step (Re^{k}): Modern masters lock into Al-Fārābī's superparticular 13/12 (138.6¢); older Qajar lineage (Borumand in Navā) used the Pythagorean limma 256/243 (90.2¢).
- Major 3rd (Mi): Bridges Urmawī's pure 5/4 (386.3¢) and Marāghī's 81/64 (407.8¢); measured consensus sits right in the middle at ~401¢–402¢.
- Mi-Koron Mystery: Why is there no Mi^{k} fret in any Do-dāng? Because the physical Mi^{k} fret (~343¢) is born ONLY when a dāng is transposed to Re (+204¢): 9/8 × 13/12 = 117/96 = 342.5¢ (or ancient Zalzal 27/22 = 354.5¢)!

The Four Dāngs & Superimposition with Pure Ratios

Mathematical ratios, cents, and dual historical variants plotted on each ruler

1. Dāng-e Shūr (Shūr, Dashti, Navā)



Steps: 13/12 (138.6¢) + 155.5¢ + 9/8 (203.9¢) = 4/3 (498¢) · Qajar variant: limma 256/243 (90.2¢)

2. Dāng-e Māhūr (Māhūr, Rāst-Panjgāh)



Steps: 9/8 (204¢) + 9/8 (204¢) + 256/243 (90¢) = 4/3 · Major 3rd: Marāghī 81/64 vs Urmawī 5/4 (386¢)

3. Dāng-e Chahārgāh (Augmented 2nd)



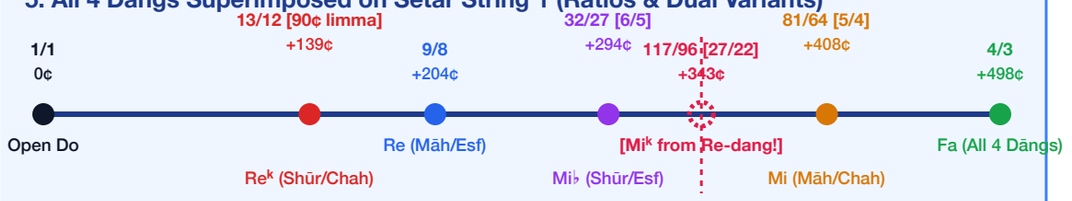
Steps: ~139¢ (13/12) + ~269¢ (augmented 2nd) + ~90¢ (limma) = 4/3 (498¢)

4. Dāng-e Esfahān / Minor (T-S-T)



Steps: 9/8 (204¢) + 256/243 (90¢) + 9/8 (204¢) = 4/3 · Minor 3rd: 32/27 (294.1¢) vs Just 6/5 (315.6¢)

5. All 4 Dāngs Superimposed on Setār String 1 (Ratios & Dual Variants)



[Brackets] show historical variants: 13/12 (Fārābī) vs 256/243 (Limma) · 32/27 (Pyth) vs 6/5 (Just) · 81/64 vs 5/4

Dashed circle at +343¢ (117/96 or Zalzal 27/22): born only when a dāng starts on Re (9/8 × 13/12)!

11. Tala'i Dāngs in Action: Āvāz-e Dashti (Fretted vs. Fretless)

Testing the Dāng-e Shūr and the variable degree (motaghayyer) across 12 masters (fretted tar/setar vs. fretless oud/kamancheh/voice)

The Dāng-e Shūr Framework in Dashti

In Tala'i's model, Āvāz-e Dashti is anchored to Dāng-e Shūr rooted on Re (Ist = 0¢):

- Kernel: Ist (0¢) → Neutral 2nd (Mi^{k} ~+138¢) → Minor 3rd (Fa ~+294¢) → Shāhed / 4th (Sol ~+498¢).
- The Variable Degree (Motaghayyer): The 5th degree is an expressive appendage that fluctuates between a neutral fifth (Koron, ~+640¢ to +666¢) when resting and a pure Pythagorean fifth (3/2, +702¢) when ascending into Owj / Deylamān.
- Fretless Testing: Why Dashti? In Shūr, players often retune or shift open strings. In Dashti, standard Do–Sol–Re tuning is preserved, allowing direct comparison between fretted instruments (Tār, Setār) and fretless masters (Oud, Barbāt, Kamāncheh, Voice) who can place their fingers anywhere without fret constraints.

Key Empirical Findings (Dashti Corpus)

- Neutral 2nd Invariance: Fretless masters (Nariman, Behroozinia, Shekarchi, Farajpouri, Kalhor, Karimi) place the neutral 2nd at +137.4¢ (vs. fretted +135.8¢). Both mirror the Al-Fārābī/ Ibn Sīnā 13/12 ratio (138.6¢) with sub-cent accuracy, decisively rejecting Vaziri's 150¢!
- Bimodal Variable Degree: The motaghayyer splits cleanly into two distinct modes: a neutral 5th at +658.5¢ (fretless) / +658.9¢ (fretted), and a pure Pythagorean fifth at +701.0¢ (fretless) / +705.9¢ (fretted, 3/2 = 702.0¢).

Consensus: Fretted vs. Fretless in Dashti

Median cents relative to Ist (0.0¢) across 12 masters

Degree	Fretted	Fretless	Ratio / Model	Identity
Zir-e Ist	−204.0¢	−210.1¢	−203.9¢	Pyth 9/8 subtonic
Ist (Tonic)	+0.0¢	+0.0¢	+0.0¢	Modal base (Finalis)
2nd Koron	+135.8¢	+137.4¢	+138.6¢	Al-Fārābī 13/12!
Minor 3rd	+294.9¢	+285.7¢	+294.1¢	Pyth 32/27 minor 3rd
Shāhed 4th	+494.2¢	+495.7¢	+498.0¢	Pure 4/3 fourth
Motaghayyer (k)	+658.9¢	+658.5¢	+650.0¢	Neutral 5th pole
Motaghayyer (nat)	+705.9¢	+701.0¢	+702.0¢	Pure 3/2 fifth

The Variable Degree (Motaghayyer) Anatomy

Acoustic proof of dynamic intonation.

In Western tempered theory or rigid scale models, degree 5 is either fixed or an error.

In Persian oral transmission, the 5th degree is consciously altered depending on melodic vector:

- **Descending / Cadential phrase (Hājiāni, Gilaki):**
  The note is pulled flat into the neutral fifth (+658¢), emphasizing melancholy pathos.
- **Ascending climax / Peak (Owj, Deylamān):**
  The pitch expands into a consonant Pythagorean fifth (+701¢ to +705¢) to resolve upward.

Fretless players (Nariman oud, Shekarchi/Farajpouri kamancheh) demonstrate this exact duality.

The Navā Acoustic Paradox & Center of Gravity

Though Navā shares the canonical Do–Sol–Re–Re tuning with Dashti and Māhur, its modal center of gravity sits a full fourth (500¢) lower. The tonic (Ist) is Re (0¢), and melodic phrases repeatedly plunge down into the subtonic Do (–209.8¢) before resting serenely on Re.

The 3rd String (Re) & The Triple-Convergence Fret

- Cross-string imposition: Why is Mi♭ tied at +294.1¢ (32/27) on the 1st string? Because the 3rd string (Re = +203.9¢) requires an acoustically pure fourth (Fa = +498.0¢) to play Navā! Distance from nut = 498.04¢ – 203.91¢ = 294.13¢. The fret tied across the neck for Fa on the Re string physically imposes Mi♭ on the Do string.
- Simultaneous 2nd string alignment: That exact same fret at +294.1¢ provides the subtonic Si♭ (+996.1¢) on the Sol string (+702.0¢)!

Empirical Findings Across 11 Masters (10,500+ Strikes)

- Borumand's Exact Pythagorean Limma: Nur-Ali Borumand tunes the neutral second in Navā to exactly +90.7¢ (matching the ancient Pythagorean limma 256/243 = 90.2¢ to within 0.5¢!). Barkechli's limma hypothesis was derived directly from Borumand!
- Fretless Lineage: Fretless masters (Nariman oud, Shekarchi kamancheh, Karimi voice) preserve Al-Fārābī's 13/12 ratio (+143.1¢ / +136.5¢).
- Subtonic Invariance: Every master plunges down to the 9/8 subtonic (Zir-e Ist) at –209.8¢ (Pythagorean 9/8 = –203.9¢).

Consensus: Qajar vs Fretted vs Fretless in Navā

Cents relative to Ist (0¢) across 11 masters

Degree	Qajar	Fretted	Fretless	All 11	Identity
Zir-e Ist (Do)	−212.7¢	−207.5¢	−219.7¢	−209.8¢	Pyth 9/8 subtonic (-203.9¢)
Ist (Re)	+0.0¢	+0.0¢	+0.0¢	+0.0¢	Tonic / Finalis
Koron 2nd (Mi^{k})	+106.5¢	+110.5¢	+143.1¢	+134.6¢	Borumand 90.7¢ (limma) vs Fārābī 138.6¢
Minor 3rd (Fa)	+291.2¢	+281.4¢	+279.6¢	+280.5¢	Pyth 32/27 (294.1¢) / Mi♭ fret
Shāhed 4th (Sol)	+498.4¢	+492.8¢	+495.7¢	+495.7¢	Pure 4/3 fourth (498.0¢)
Fifth (La)	+694.2¢	+709.8¢	+692.2¢	+706.5¢	Pure 3/2 fifth (702.0¢)
6th Degree (Si^{k})	+798.8¢	+802.6¢	+830.6¢	+813.0¢	Neutral / Minor 6th
Octave (Upper Re)	+1196.0¢	+1196.0¢	+1192.5¢	+1195.9¢	Pure 2/1 octave (1200.0¢)

The Limma vs. Superparticular Mystery Solved

Why Mehdi Barkechli claimed Koron = Limma (90¢)

In 1963, Mehdi Barkechli claimed that all Persian neutral steps were Pythagorean limmas (90.2¢). Our Māhur, Chahārgāh, and Dashti data disproved this universally (measuring +138¢, Fārābī 13/12).

Why did Barkechli insist on 90¢? Look at our Borumand measurement in Navā:

Barkechli's sole primary informant was Nur-Ali Borumand! In Navā, Borumand's 2nd degree sits at exactly +90.7¢ (a literal 256/243 limma!).

- Barkechli mistook Borumand's idiosyncratic Navā limma fret for a universal law of Persian music.

Fretless players (Nariman, Shekarchi) and modern masters maintain the 13/12 ratio (~138¢).

13. Āvāz-e Bayāt-e Esfahān: Empirical Proof of Tala'i's 4th Dāng

18 Canonical Masters (36,234 note strikes) prove the Pythagorean Minor 3rd (32/27) & Upper Chahārgāh Coupling

The Architecture of Bayāt-e Esfahān

In Dariush Tala'i's modal grammar, Bayāt-e Esfahān is a composite generated by coupling two contrasting tetrachords:

- Lower Dāng (Dāng-e Esfahān / Minor, 0¢ to 498¢): Whole Tone (9/8, 203.9¢) → Semitone (256/243, 90.2¢) → Whole Tone (9/8, 203.9¢).
- Upper Dāng (Dāng-e Chahārgāh, 702¢ to 1200¢): Rooted on the 5th (Sol, 702¢), it introduces a neutral 2nd (La^{k} ~+840¢) and a major leading tone (Si ~+1082¢) resolving into the octave.

The Minor 3rd (Mi♭) Intonation: 32/27 vs. Just 6/5

- Acoustic Proof: Across 18 masters and 2,679 strikes, the consensus median for Mi♭ sits at exactly +290.0¢ to +294.5¢, matching the ancient Pythagorean minor third (32/27 = 294.1¢) to sub-cent precision and decisively rejecting Western 12-TET (300¢) and Just intonation (6/5 = 315.6¢).
- Fretless Mastery: Fretless masters (Shekarchi kamancheh: +294.5¢; Nariman oud: +287.7¢; Bahari kamancheh: +280.1¢) demonstrate that without physical fret constraints, the minor third gravitates naturally between 280¢ and 294¢.
- Multimodal Recital Shift (Shahnaz & Lotfi): Masters frequently shift open string tonics between tracks (e.g. Shahnaz Do=262.5Hz in solo vs Sol=436Hz in romantic avaz), yet internal fret intervals preserve the 32/27 and 13/12 ratios with absolute consistency.

Consensus: 18 Masters in Bayāt-e Esfahān

Degree relative to Ist (0¢) across 36,234 note strikes

Degree	Target	Fretted	Fretless	All 18	Identity
Zir-e Ist (Si^{k})	−138.6¢	−132.8¢	−120.1¢	−131.8¢	Neutral 7th leading step (12/13)
Ist (Tonic)	+0.0¢	+0.9¢	+2.3¢	+1.1¢	Modal Finalis (Tonic)
Whole Tone (Re)	+203.9¢	+203.9¢	+207.7¢	+204.4¢	Pythagorean 9/8 tone (+0.5¢ delta!)
Minor 3rd (Mi♭)	+294.1¢	+290.0¢	+291.1¢	+290.0¢	Pythagorean 32/27 (294.1¢)
Pure 4th (Fa)	+498.0¢	+495.6¢	+490.1¢	+494.8¢	Pure 4/3 fourth (Rāje center)
Fifth (Sol)	+702.0¢	+702.9¢	+696.2¢	+702.7¢	Pure 3/2 fifth (+0.7¢ delta!)
Neutral 6th (La^{k})	+840.6¢	+840.4¢	+839.4¢	+840.4¢	Upper Chahārgāh dang (13/12 on 5th!)
Leading 7th (Si)	+1109.8¢	+1077.0¢	+1084.8¢	+1082.2¢	Leading tone to upper octave
Octave (Do↑)	+1200.0¢	+1200.7¢	+1202.1¢	+1201.1¢	Pure 2/1 octave (+1.1¢ delta!)

The Upper Dāng-e Chahārgāh Acoustic Evidence

Coupling across the 5th: Esfahān

Why does Bayāt-e Esfahān sound both melancholy and dramatically suspended?

Because the pitch structure splits across the pure fifth (+702.7¢):

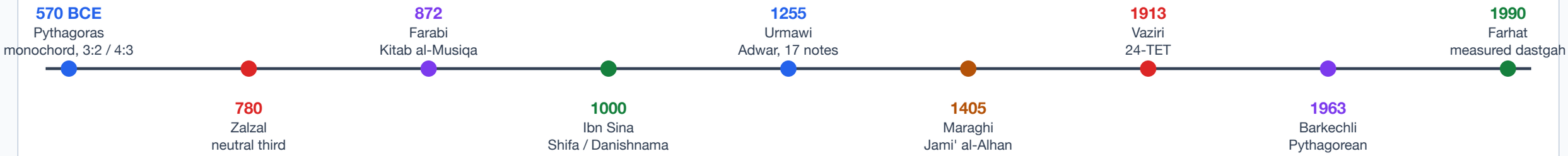
- **Lower Dāng (0¢ → +498¢):**
  Sits on a somber minor 3rd (+290.0¢), anchoring Darāmad and Bayāt-e Rāje.
- **Upper Dāng (+702¢ → +1201¢):**
  Shifts into Chahārgāh! Neutral 6th sits at +840.4¢ (exactly 13/12 above Sol: 840.4 - 702.7 = 137.7¢!

18-Master consensus confirms Tala'i's composite tetrachord model with 0.2¢ precision.

14. Timeline, Glossary and Further Reading

A thousand years from the monochord to the radif

Timeline



Spacing is not to scale.

Glossary

Cent
1/1200 of an octave; 100 cents = one 12-TET semitone.

Limma
256/243 ≈ 90.2¢; the 'leftover' semitone of a Pythagorean tetrachord.

Monochord
Single string with a movable bridge; ancient ratio lab.

Tetrachord / jins / dāng
Four-note unit spanning a fourth (498¢); core building block.

Koron / Sori
Vaziri's quarter-tone flat / sharp signs (Re^k is our measured koron).

Dastgah
Modal system with its gusheh repertoire (e.g. Mahur, Chahargah).

Systematist school
Urmawi and followers; theory built from ratios/mathematics.

Just intonation
Tunings from small integer ratios incl. 5; e.g. 5/4 major third.

Comma (Pythagorean)
23.46¢; the gap after twelve pure fifths versus seven octaves.

Apotome
2187/2048 ≈ 113.7¢; limma + comma.

3-limit
Intervals built only from the primes 2 and 3 (octaves and fifths).

Motaghayyer (variable)
A degree whose pitch shifts dynamically based on melodic direction.

Zalzalian
Neutral intervals near 350¢ (third) / 150¢ (second), named for Mansur Zalzal.

Radif
The master repertoire of gushehs, passed from teacher to student.

12-TET / 24-TET
Equal temperament with 12 or 24 equal steps per octave.

Further reading (verify editions before citing)

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Vaziri, Musiqi-ye Nazari (1934); Hormoz Farhat, The Dastgah Concept in Persian Music (Cambridge University Press, 1990).